

I wrote another meticulously sourced article about beef vs AI water usage for everyone to ignore. Maybe I should find a way to post it on the tiktok? Would it be more convincing if I provided a google docs revision history where I say the exact same thing written manually without AI assistance?

Posted by u/ram_altman • • 0 points • 10 comments

If you used ChatGPT 50 times a day, every single day, for an entire year, you could offset all of that water consumption by skipping one or two hamburgers. That's the actual math.

A chart has been going around Reddit comparing AI's water footprint to a hamburger's. It claims 300 ChatGPT queries use about 1 gallon of water and one hamburger uses 660 gallons. Both numbers are wrong.

The 660 gallon burger figure comes from the Water Footprint Network via the documentary Cowspiracy. It represents the total water footprint of a quarter-pound of beef, meaning it bundles together green, blue, and grey water into one number [1]. Green water is just rain. It falls on pastures and feed fields and gets taken up by plants. It would fall there regardless of whether anyone was raising cattle. Blue water is freshwater actively pulled from rivers, lakes, and aquifers, the stuff that comes out of taps and competes with drinking water and ecosystems. Grey water is a theoretical volume representing how much freshwater you'd need to dilute pollutants to safe levels. Over 90 percent of beef's total water footprint is green water [4]. Including it makes beef look enormously water-intensive, but rain falling on a pasture in Missouri is not the same thing as pumping the Ogallala Aquifer.

The AI number on the chart, meanwhile, counts only blue water. So the comparison is broken from the start.

What does a burger actually cost in blue water? The most cited U.S. study is Beckett and Oltjen 1993, published in the Journal of Animal Science. They excluded all rainfall and counted only irrigation, livestock drinking water, and processing water. They got 441 gallons per pound of boneless beef [2]. A 2022 update by Klopatek and Oltjen using 2019 USDA data found that number had dropped 37.6 percent to about 275 gallons per pound, thanks to better irrigation, higher crop yields, and more byproducts in feedlot rations [3]. Kansas

State's Beef Cattle Institute, using a different methodology, put the combined blue and grey water at 158 gallons per pound [4].

Some caveats on these numbers. The Beckett and Oltjen line of research has industry connections. The original 1993 study was partly funded by the California Beef Council, and the National Cattlemen's Beef Association has used their figures for decades to counter higher estimates from environmentalists. The 2022 update was co-authored by Oltjen himself [5]. That doesn't invalidate the work, and at least one independent analysis concluded it's probably the best blue water estimate for U.S. beef [5], but you should know who's behind it. Also, both studies measure water withdrawals, not consumption. They count all irrigation water applied but don't subtract the 10 to 20 percent that runs off and returns to the water supply [5]. The actual consumptive blue water is somewhat lower.

So the honest range for a quarter-pound burger is roughly 40 to 70 gallons of blue water, depending on methodology and assumptions. Wide range, real uncertainty, but a fraction of the 660 on the chart.

Now the AI side. The "500ml per conversation" figure everyone cites comes from a 2023 paper by Li, Ren, and others, but its power estimates were based on GPT-3 data from 2020 [6]. Models have gotten dramatically more efficient since then.

Google's August 2025 technical report measured the median Gemini text prompt at 0.26 milliliters of water [7]. Sam Altman said in June 2025 that the average ChatGPT query uses about 0.3 milliliters [8]. An independent benchmarking study measured a short GPT-4o query at roughly 0.5 to 0.8 milliliters [9]. Google's figure only covers on-site cooling. Including off-site electricity water brings it to probably 1.5 to 3 milliliters per prompt [10]. Google's report has also been criticized for using median instead of mean, not specifying prompt length, and using market-based carbon accounting [11]. Longer queries cost much more. A medium-length GPT-5 response has been estimated at 25 to 39 milliliters [12].

There's also the stuff that happens before you ever type a prompt. Training GPT-4 consumed 11.5 to 13.4 million gallons of water per month at Microsoft's Iowa data centers during peak intensity in 2022 [13]. Amortized across hundreds of billions of queries over the model's life, that adds maybe 0.1 to 0.5 milliliters per query. Chip manufacturing takes about 2,200 gallons of water per silicon wafer [14], with TSMC alone consuming 101 million cubic meters in 2023 [15], but each GPU serves millions of queries over years, so per query it's fractions of a milliliter.

Add it all up: inference, off-site electricity, amortized training, amortized silicon. A reasonable full-lifecycle estimate for a typical short query on a current model is about 3 to 10 milliliters. All blue water.

Say you're a heavy user. 50 queries a day, every day, all year. At 3 to 10 milliliters per query, that's 150 to 500 milliliters of blue water per day. Over a year, roughly 55 to 180 liters, or 14 to 48 gallons.

One quarter-pound hamburger costs 40 to 70 gallons of blue water.

Your entire year of heavy AI use costs less water than one or two burgers. Even using the most aggressive AI estimates and the most conservative beef numbers, you're talking about skipping maybe three or four burgers across a whole year to break even.

The average American eats about 57 pounds of beef per year. At 158 to 275 gallons of blue water per pound, that's roughly 9,000 to 15,700 gallons of blue water just from beef annually. A heavy AI user's annual water footprint of 14 to 48 gallons is a rounding error on that.

At the aggregate level the picture is similar. The Water Footprint Network estimates that global beef production uses about 800 to 900 cubic kilometers of water per year across all water types [16]. The World Economic Forum puts the total global AI economy at about 23 cubic kilometers [17]. Beef is roughly 35 to 40 times larger globally, and that's comparing beef's total footprint (green + blue + grey) against AI's mostly-blue footprint. If you could isolate global beef blue water alone it would shrink, but it would still dwarf AI by a wide margin.

And the trajectory for AI water use is actually improving, not just per query but at the infrastructure level. Most of today's data center water consumption comes from evaporative cooling, where water absorbs heat from servers and then evaporates in cooling towers, lost to the atmosphere. It works the same way your body cools itself by sweating. But the industry is moving away from this.

Closed-loop cooling systems recirculate coolant without evaporating it, cutting freshwater consumption dramatically. Brookings estimates that closed-loop systems can reduce freshwater use by up to 70 percent [18]. Liquid immersion cooling, where servers are submerged in non-conductive fluid, can cut water consumption by up to 91 percent compared to conventional air cooling [19]. Direct-to-chip cooling, which runs coolant directly across processor surfaces, can reduce water use by 20 to 90 percent depending on climate and system design [19].

Oracle announced in early 2026 that its new AI data centers in New Mexico, Michigan, Texas, and Wisconsin will deploy closed-loop cooling that does not rely on continuous consumption of potable water [20]. Microsoft stated that starting August 2024, all new datacenter designs use next-generation cooling technology aimed at zero water evaporation, with the first sites coming online in late 2027 [21]. Edged US broke ground on a facility in Aurora, Illinois designed to save more than 277 million gallons of water annually

compared to conventional evaporative approaches [22].

There is a catch. Some of the most promising liquid cooling technologies use fluorinated fluids that fall under the umbrella of PFAS, the "forever chemicals" that are increasingly regulated. That has made some companies cautious about adoption [23]. And closed-loop systems trade water consumption for higher electricity use, which has its own environmental footprint. It's not a free lunch. But the direction is clear: the water-per-query number, already small, is heading toward near zero for on-site consumption at new facilities.

But the next time someone tells you that using ChatGPT is an environmental sin, ask them if they ate a burger this week. If they did, that one meal used more water than your AI habit will all year. The moral panic about AI water use isn't rooted in the numbers. And the people who should actually be scrutinized are not individual users typing questions into a chatbot. They're the companies deciding where to build data centers, what cooling systems to install, and how much of their actual consumption to disclose.

Sources:

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[2] Beckett, J.L. and J.W. Oltjen. "Estimation of the water requirement for beef production in the United States." Journal of Animal Science, 71(4): 818-826. 1993.

[3] Klopatek, S.C. and J.W. Oltjen. "How advances in animal efficiency and management have affected beef cattle's water intensity in the United States: 1991 compared to 2019." Journal of Animal Science, 100(11). 2022.

[4] Lancaster, P. "Does beef production really use that much water?" Kansas State University Beef Cattle Institute. 2020.

[5] "The Fine Print on Beef's Water Use." reducing-suffering.org. Analysis of Beckett and Oltjen methodology, industry funding, and withdrawal vs. consumption distinction.

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[7] Google Cloud Blog. "Measuring the environmental impact of AI inference." August 2025.

[8] Altman, S. OpenAI blog post. June 2025.

[9] "How Hungry is AI? Benchmarking Energy, Water, and Carbon Footprint of LLM Inference." arXiv:2505.09598. May 2025.

[10] Masley, A. "An example of what I consider a misleading article about AI and the environment." Blog post. August 2025.

[11] "Is Google's Reveal of Gemini's Impact Progress or Greenwashing?" Towards Data Science. August 2025.

[12] Lo, L.S. "AI has a hidden water cost: here's how to calculate yours." The Conversation. 2025.

[13] "How Much Water Does AI Use? The Real Numbers for 2026." AI Tool Discovery. March 2026.

[14] "8 Things You Should Know About Water & Semiconductors." Center for Water Research and Resilience.

[15] "Water Usage in Semiconductor Manufacturing to Double by 2035." IDTechEx Research. March 2025.

[16] Water Footprint Network. Global water footprint of animal production, citing Hoekstra 2012 and Mekonnen and Hoekstra 2012.

[17] World Economic Forum. "Why AI's water problem might actually be an opportunity." January 2026.

[18] Brookings Institution. "AI, data centers, and water." November 2025.

[19] World Economic Forum. "What new water circularity can look like for data centres." November 2025.

[20] Oracle Cloud Infrastructure. "Closed-loop cooling in Oracle AI data centers." February 2026.

[21] Microsoft Cloud Blog. "Sustainable by design: Next-generation datacenters consume zero water for cooling." December 2024.

[22] Data Centre Magazine. "How Closed-Loop Cooling Is Reshaping Data Centre Design." February 2026.

Comments

u/zebrasmack • 1 points • 2026-03-30 13:38 UTC

so your argument is these companies *plan* to make *new* data centers better at water consumption? why didn't you just include the current ones and current impact? with a note of how the new plans could change things?

I do agree animal husbandry need to really focus on making cows more environmentally friendly, though.

u/ram_altman • 1 points • 2026-03-30 20:33 UTC

>why didn't you just include the current ones and current impact?

I did, you just refuse to read.

>I do agree animal husbandry need to really focus on making cows more environmentally friendly, though.

So how do you propose that?

u/nuker0S • 1 points • 2026-03-28 11:45 UTC

The question is how much water can be saved by eating something else than a burger while providing the same calories

u/SweetCommieTears • 7 points • 2026-03-28 08:25 UTC

Have you ever heard the saying

> "you cannot reason a person out of a position he did not reason himself into in the first place"?

Or

>"Never argue with stupid people, they will drag you down to their level and then beat you with experience."

Or even

>"Never argue with liars. You can't win, because they believe their own lies"

If you have heard these sayings, consider that antis are not holding a rational or logical position. The logical train doesn't go "I researched and AI is especially bad to the environment, therefore I'm against AI" It goes "I hate AI. It also affects the environment, so I will use this as a weapon to argue against it".

Antis KNOW that AI doesn't affect the environment that much. They KNOW that streaming, burgers,

cheap jeans, whatnot all use more water and electricity. However, it's easy to go and leap into logic "well, actually netflix and hulu are necessary" or "well, actually food (disingenuously conflating burgers with ALL food) is necessary, AI is not".

They also KNOW that a datacenter being built near some community and making their water bills surge isn't an AI problem but an American capitalism one. However, to acknowledge this means to stand up to Big Daddy government and that one you can't brow-beat as easily.

The truth of the matter, and something every human who is grown enough to pay their own bills knows (so hardly 20% of antis), is that every single aspect of modern human existence is damaging to the environment. Much of it extremely so, and we don't really care because it's inconvenient. For AI, it's just brought up because it's a very effective piece to parrot. It allows you to demonize your opposition while also making your position appear more moral.

u/Jbern124 • 3 points • 2026-03-28 11:56 UTC

It depends on what kind of antis they are. I call myself an "anti", but I would use AI to look at color schemes, room layouts, car exterior and interior designs, and trail route mapping. The anti aspect comes into play when it's used solely for content moderation, surveillance, and grocery store pricing.

u/ReasonableWelder51 • 3 points • 2026-03-28 13:02 UTC

You are hardly an anti when you see a purpose in using AI and only disagree with the scummy purposes. The majority of pro/neutral people shares the same beliefs.

u/Full_Boysenberry_314 • -2 points • 2026-03-28 04:34 UTC

If you want something to be true you have to say it very often and loudly. Use many formats. Consider enlisting bot farm message amplification services. Traction does tend to cost money. What is your budget for this truth production project?

u/Kind-Scheme7517 • 7 points • 2026-03-28 03:32 UTC

Investing in this comment please notify me when this clearly high effort post gains traction

u/[deleted] • 1 points • 2026-03-28 03:18 UTC

[removed]

u/Ximena-WD • -4 points • 2026-03-28 03:01 UTC

Don't do that. Do use sound logic and reasoning to make ai sound better

u/Kind-Scheme7517 • 6 points • 2026-03-28 03:31 UTC

Is this a grammar issue or just stupid?

u/Ximena-WD • 2 points • 2026-03-28 03:43 UTC

Yes